Mahmud Sani Madobi

AI Systems Engineer and Software Architect · Founder and CEO, MeerTech Ltd

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PROFILE

I design, build and run production systems for energy, finance and data, for places where the power, the network and the inputs are unreliable. I work at the systems layer: double-entry ledgers with idempotent writes, offline-first sync, row-level security, and AI kept inside deterministic boundaries. As founder of MeerTech Ltd I am responsible for the architecture of its products and client systems and for keeping them running.

EXPERIENCE

Founder and CEO, MeerTech Ltd Present
Founded and leads an engineering company that builds and operates software for energy, agriculture, trade, finance and the public sector. Designs the architecture of its products and client systems and runs them in production.

Software Engineer, part-time, a solar mini-grid developer in Nigeria 2022 to 2026
Software engineering for the operations of a solar mini-grid developer.

Digitalisation Consultant, contract, GIZ Nigeria and ECOWAS 2025
Digital maturity assessments and hands-on tool setup for small businesses on a digital transformation programme.

NYSC Intern, 5G Team, Huawei 2025
National service year placement in the Service Experience Department.

Backend and AI Engineer, trainee, MIDS Dynamics Ltd 2024 to 2025
Backend and AI engineering as a trainee.

Student Intern, Blue Sapphire Hub 2023 to 2024
Internship at a technology hub.

SELECTED SYSTEMS

RanaWallet (production). A wallet for buying mini-grid electricity over whatever channel a customer has, with a double-entry ledger underneath and settlement for the operators. FastAPI, PostgreSQL, Redis, Next.js, React.

ClearMind (production). Tasks, projects, notes, habits and goals in one offline-first workspace, opened to AI agents through an MCP server with scoped OAuth access. React, TypeScript, Vite, Firebase, Expo.

ExportFlow (production). Reads export documents, checks them against each other and against destination requirements, and grades every finding before a mismatch holds a shipment at the port. Python, FastAPI, PostgreSQL, Tesseract OCR, ClamAV.

Agricultural intelligence platform (production). A real-time GIS platform: farm and cooperative records, satellite and vegetation layers, crop suitability, and pest and disease risk advisories in the farmer's language. Next.js, PostgreSQL, PostGIS, Prisma, MapLibre.

SalCollect (production). Offline-first field verification: GPS-tagged photo evidence, six harmonised verification instruments and a tamper-evident audit chain. Expo, React Native, TypeScript, SQLCipher, Next.js.

Mini-grid remote monitoring (production). A dashboard that pulls live data from two vendors' equipment at solar mini-grids, flags sites whose data has gone stale, and logs every remote control action. Node.js, TypeScript, InfluxDB, MQTT, React.

NMAS (production). Analytics infrastructure that turns music streaming and platform activity into quarterly indicators, with methodology and lineage the agency can defend. Python, FastAPI, SQLModel, PostgreSQL, React.

WisdomSigma CIU `(production)`. A phone app that stands in for a prepayment meter's keypad unit: it sends recharge tokens over encrypted Bluetooth, reads meter values and works fully offline. Expo, React Native, TypeScript, SQLite, Bluetooth Low Energy.

Remote Portal `(production)`. Projects, documents, approvals and team chat for a mini-grid developer, with site progress reports that produce a slide deck in the company's own template. React, Vite, Supabase, FastAPI, python-pptx.

Statuses are stated from evidence. Architecture and decisions for each: sm.meertech.tech/work

TECHNICAL STACK

AI and machine learning	Python, TensorFlow and Keras, Model Context Protocol, Gemini API, OCR and document extraction, Agent safety gates
Backend	FastAPI, Node.js and TypeScript, REST APIs, WebSockets, USSD and SMS
Data	PostgreSQL, PostGIS, TimescaleDB, InfluxDB, SQLite and SQLCipher, Redis, SQLAlchemy and Alembic, Prisma
Frontend and mobile	React and Next.js, React Native and Expo, TypeScript, Kotlin (native modules), MapLibre and deck.gl
Infrastructure and IoT	Docker, nginx, MQTT, DLMS/COSEM over Bluetooth, GitHub Actions
Security	Row-level security, Schema per tenant, Idempotency keys, Append-only audit and hash chains, AES-GCM, OAuth scoped tokens

EDUCATION

MSc Financial Engineering, WorldQuant University
`In progress`

BSc Computer Science and Information Technology, Federal University Dutsin-Ma `2024`

Software Engineering programme, ALX Africa `2025`

Applied Data Science Lab, WorldQuant University `2025`

Aspire Leaders Program, Aspire Institute `2025`

CREDENTIALS AND MEMBERSHIPS

HCIP-AI, Huawei `2025`

Certified Backend Engineer, ALX Africa `2025`

Member, Association for the Advancement of Artificial Intelligence (AAAI) `2026`

Member, Association for Computing Machinery (ACM) `2026`

Global member, Internet Society `2026`

Vice President, Student Union Government, Federal University Dutsin-Ma `2023 to 2024`

LANGUAGES

English and Hausa.